

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401311
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Seo; Yuichi et al.

Motor control device

Abstract

A motor control device includes: a rotation control determination unit configured to determine whether a motor rotation control to rotate the motor fails; a parameter calculation unit configured to calculate a control failure frequency parameter having a correlation with a frequency of failure in the motor rotation control based on a determination result by the rotation control determination unit; an abnormality determination unit configured to determine whether an abnormality has occurred in the motor based on the control failure frequency parameter; and a stop suppression unit configured to suppress stop of rotation of the motor by changing a motor control parameter to perform the motor rotation control when the abnormality determination unit determines that an abnormality has occurred in the motor.

Inventors:	Seo; Yuichi (Kariya, JP), Hidaka; Yuuji (Kariya, JP)
Applicant:	DENSO CORPORATION (Kariya, JP)
Family ID:	1000008781576
Assignee:	DENSO CORPORATION (Kariya, JP)
Appl. No.:	18/324222
Filed:	May 26, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240048084 A1	Feb. 08, 2024

Foreign Application Priority Data

JP	2022-123989	Aug. 03, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: H02P29/028 (20160101); F02D41/22 (20060101); F04D27/00 (20060101); G07C5/08 (20060101); G08B21/18 (20060101)

U.S. Cl.:

CPC H02P29/028 (20130101); F02D41/221 (20130101); G07C5/0808 (20130101); G08B21/187 (20130101); F02D2041/224 (20130101); F04D27/001 (20130101)

Field of Classification Search

CPC: H02P (29/028); F02D (41/221); G07C (5/0808); G08B (21/187)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2002/0147536	12/2001	Sugita	701/62	F16H 57/01
2013/0193894	12/2012	Kiguchi	318/490	H02P 29/024
2013/0238147	12/2012	Okamoto	700/282	H02P 29/0241
2018/0234039	12/2017	Kuwahara	N/A	B62D 5/0463
2018/0274672	12/2017	Kondo	N/A	G05B 23/02
2021/0075356	12/2020	Takahashi	N/A	H02P 21/22
2022/0194542	12/2021	Kirchhoff	N/A	B63H 21/21

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2009-077456	12/2008	JP	N/A
2021-019398	12/2020	JP	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 18/324,247, filed May 26, 2023, Seo et al. cited by applicant

U.S. Appl. No. 18/324,241, filed May 26, 2023, Seo et al. cited by applicant

Primary Examiner: Dhakal; Bickey

Attorney, Agent or Firm: Posz Law Group, PLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is based on Japanese Patent Application No. 2022-123989 filed on Aug. 3, 2022, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a motor control device that controls a motor.

BACKGROUND

(3) A motor control device determines a sign of an abnormality by comparing a motor current value, a motor voltage value, a motor rotation speed, and the like with a predetermined determination threshold value when a motor is driven to function as a source for driving a fuel

pump.

SUMMARY

(4) According to an aspect of the present disclosure, a motor control device configured to control a motor includes a rotation control determination unit, a parameter calculation unit, an abnormality determination unit, and a stop suppression unit.

(5) The rotation control determination unit is configured to determine whether a motor rotation control to rotate the motor has failed. The parameter calculation unit is configured to calculate a control failure frequency parameter having a correlation with a frequency of failure in the motor rotation control based on a determination result by the rotation control determination unit.

(6) The abnormality determination unit is configured to determine whether an abnormality has occurred in the motor based on the control failure frequency parameter. The stop suppression unit is configured to control the motor by changing a motor control parameter to perform the motor rotation control so as to suppress rotation of the motor from being stopped when the abnormality determination unit determines that an abnormality has occurred in the motor.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram illustrating a configuration of a fuel supply system.

(2) FIG. 2 is a block diagram illustrating a configuration of a fuel pump and a fuel pump controller.

(3) FIG. 3 is a sectional view illustrating the fuel pump.

(4) FIG. 4 is a flowchart illustrating a motor control process according to a first embodiment.

(5) FIG. 5 is a flowchart illustrating a motor control process according to a second embodiment.

DETAILED DESCRIPTION

(6) A motor control device determines a sign of an abnormality by comparing a motor current value, a motor voltage value, a motor rotation speed, and the like with a predetermined determination threshold value when a motor is driven to function as a drive source of a fuel pump.

(7) The torque applied to the motor of the fuel pump increases when the pressure of the fuel increases, when a foreign matter is caught in the impeller of the fuel pump, or when the impeller of the fuel pump deforms and interferes with the casing of the fuel pump.

(8) As a result of detailed studies by the inventors, when a foreign substance is caught in the impeller of the fuel pump or when the impeller and the casing interfere with each other, the rotation of the motor of the fuel pump may stop and the fuel pump may not be able to pump up fuel.

(9) The present disclosure provides a motor control device to improve the reliability of a fuel pump.

(10) According to one aspect of the present disclosure, a motor control device configured to control a motor includes: a rotation control determination unit; a parameter calculation unit; an abnormality determination unit; and a stop suppression unit.

(11) The rotation control determination unit is configured to determine whether or not a motor rotation control for rotating the motor has failed. The parameter calculation unit is configured to calculate a control failure frequency parameter having a correlation with a frequency of failure in the motor rotation control based on a determination result by the rotation control determination unit.

(12) The abnormality determination unit is configured to determine whether an abnormality has occurred in the motor based on the control failure frequency parameter. The stop suppression unit is configured to control the motor by changing a motor control parameter for performing the motor rotation control so as to suppress rotation of the motor from being stopped when the abnormality determination unit determines that an abnormality has occurred in the motor.

(13) The motor control device of the present disclosure can determine that a foreign matter is

caught in the impeller of the fuel pump or the impeller of the fuel pump and the casing interfere with each other as abnormality when the frequency of failures in the motor rotation control increases.

(14) Therefore, the motor control device according to the present disclosure can take appropriate measures to restrict the rotation of the motor from being stopped when the abnormality occurs. Accordingly, the motor control device according to the present disclosure can restrict the rotation of the motor from being stopped when the abnormality occurs, and can improve the reliability of the fuel pump.

First Embodiment

(15) Hereinafter, a first embodiment of the present disclosure will be described with reference to the drawings. The fuel supply system **1** of the present embodiment is mounted on a vehicle. As shown in FIG. **1**, the fuel supply system **1** includes a fuel tank **2**, a fuel pump **3**, a suction filter **4**, a fuel pipe **5**, a pressure sensor **6**, an engine controller **7**, and a fuel pump controller **8**.

(16) The fuel tank **2** stores fuel to be supplied to the engine EG of the vehicle. The engine EG includes multiple injectors respectively corresponding to the multiple cylinders. The injectors inject fuel into the cylinders respectively.

(17) The fuel pump **3** is installed inside the fuel tank **2** and pumps up the fuel stored in the fuel tank **2**. The suction filter **4** is installed near the suction hole **45** of the fuel pump **3** in the fuel tank **2** and removes foreign matters from the fuel sucked by the fuel pump **3** by collecting foreign matters in the fuel.

(18) The fuel pipe **5** supplies the fuel from the fuel pump **3** to the engine EG. The pressure sensor **6** detects the pressure of fuel flowing through the fuel pipe **5** and outputs a pressure detection signal indicating the detection result.

(19) The engine controller **7** drives the injectors to control fuel injection to the engine EG. The engine controller **7** controls the fuel pump **3** via the fuel pump controller **8** so that the fuel pressure indicated by the pressure detection signal acquired from the pressure sensor **6** matches the target fuel pressure.

(20) The fuel pump controller **8** controls the fuel pump **3** based on a command from the engine controller **7**. As shown in FIG. **2**, the fuel pump **3** has a pump motor **22**. In the present embodiment, the pump motor **22** is a three-phase brushless motor.

(21) The fuel pump controller **8** includes an inverter circuit **11**, a drive unit **12** and a control unit **13**. The inverter circuit **11** receives power supply from a battery (not shown) and applies a battery voltage VB between the terminals TU, TV, TW of the pump motor **22** (between U-phase and V-phase, between V-phase and W-phase, and between W-phase and U-phase) to energize the stator coil so as to rotate the pump motor **22**.

(22) The U-phase, V-phase and W-phase stator coils of the pump motor **22** are connected in a Y-connection. The inverter circuit **11** is connected to the three terminals TU, TV, TW opposite to the Y-connection. The inverter circuit **11** includes a three-phase full-bridge circuit having six switching elements Q1, Q2, Q3, Q4, Q5, and Q6.

(23) The switching elements Q1, Q2, and Q3 are disposed as so-called high-side switches between the positive electrode side of the battery and the terminals TU, TV, and TW of the pump motor **22**. The switching elements Q4, Q5, and Q6 are disposed as so-called low-side switches between the negative electrode side of the battery and the terminals TU, TV, and TW of the pump motor **22**.

(24) Therefore, in the inverter circuit **11**, the battery voltage VB is applied between any of the terminals TU, TV, and TW of the pump motor **22** by turning on one high-side switch and one low-side switch having different phases.

(25) The terminal to which the battery voltage VB is applied and the application direction of the battery voltage VB can be switched by switching the switching element to be turned on. The current flowing through the pump motor **22** can be controlled by controlling the ON time of the switching element.

(26) The drive unit **12** turns on or off the switching elements **Q1** to **Q6** in the inverter circuit **11** in accordance with the control signal output from the control unit **13**. Thus, current flows through the U, V, W phase stator coil of the pump motor **22**, thereby rotating the pump motor **22**.

(27) The control unit **13** is an electronic control unit including as a main component a microcontroller having a CPU **13a**, a ROM **13b**, a RAM **13c**, and the like. Various functions of the microcontroller are realized by causing the CPU **13a** to execute programs stored in a non-transitory tangible storage medium. The ROM **13b** corresponds to a non-transitory tangible storage medium in which the programs are stored. A method corresponding to the program is executed by executing the program. Note that a part or all of the functions to be executed by the CPU **13a** may be configured as hardware by one or multiple ICs or the like. The number of microcontrollers configuring the control unit **13** may be one or multiple.

(28) The control unit **13** controls the current flowing through the U, V, W phase stator coil so that the target rotation speed instructed by the engine controller **7** matches the rotation speed of the pump motor **22** (the motor rotation speed). The target rotation speed is set so that the pressure of the fuel flowing in the fuel pipe **5** becomes a predetermined pressure.

(29) The fuel pump controller **8** further includes a voltage detector **14** and a current detection unit **15**. The voltage detector **14** detects the voltage V_u , V_v , V_w of the terminal TU, TV, TW of the pump motor **22**. The current detection unit **15** detects the current I_u , I_v , I_w flowing through the U, V, W phase stator coil.

(30) The detection signal of the voltage detector **14** and the detection signal of the current detection unit **15** are input to the control unit **13** and used for controlling the pump motor **22** and detecting abnormality. The control unit **13** turns on one high-side switch and one low-side switch having different phases in order to rotate the pump motor **22**. In the present embodiment, the control unit **13** rotates the pump motor **22** by performing pulse width modulation control (hereinafter, PWM control). Specifically, for example, the control unit **13** maintains one of the two switching elements to be turned on in the on state, and periodically switches the other switching element between the on state and the off state in accordance with the duty.

(31) In order to rotate the pump motor **22**, the control unit **13** switches the switching element to be turned on in synchronization with the rotational position of the pump motor **22**. In order to control the drive unit **12** in synchronization with the rotational position of the pump motor **22**, the control unit **13** detects the rotational position of the pump motor **22**. Specifically, the control unit **13** detects the rotational position of the pump motor **22** based on the voltage V_u , V_v , V_w acquired from the voltage detector **14**. The control unit **13** generates a drive command based on the detected rotational position and outputs the drive command to the drive unit **12**. Thus, the control unit **13** can control the pump motor **22** in synchronization with the rotational position of the pump motor **22**.

(32) As shown in FIG. 3, the fuel pump **3** includes a pump housing **21**, a pump motor **22**, an impeller **23**, a pump case **24**, and a motor cover **25**. The pump housing **21** is a metal member formed in a cylindrical shape.

(33) The pump motor **22** includes a rotor **31**, a stator **32**, and a shaft **33**. The rotor **31** includes a cylindrical iron core and plural pairs of magnetic poles. A permanent magnet is used for the pair of magnetic poles. The pair of magnetic poles are arranged so that the N poles and the S poles are alternately and uniformly arranged on the outer periphery of the iron core.

(34) The stator **32** is disposed at equal angular intervals around the rotor **31**, and the winding **35** is wound around the stator **32**. A U-phase, V-phase, or W-phase winding **35** is wound around the stator **32**. The shaft **33** is a metal member formed in an elongated cylindrical shape. The shaft **33** is fixed to the rotor **31** such that its axis coincides with the axis of the rotor **31**.

(35) The pump motor **22** is installed in the pump housing **21** such that the axis of the shaft **33** coincides with the cylindrical axis of the pump housing **21**. The impeller **23** is a resin member formed in a disk shape. Blade grooves **37** are arranged on the outer periphery of the impeller **23** in

the circumferential direction. The impeller 23 is fixed to the shaft 33 such that the axis thereof and the axis of the shaft 33 coincide with each other, and is disposed inside the pump housing 21 at a first end of the pump housing 21 formed in a cylindrical shape along the axial direction.

(36) The pump case 24 includes a first casing 41 and a second casing 42. The first casing 41 is installed to close the opening of the pump housing 21 at the first end of the pump housing 21.

(37) The second casing 42 is installed inside the pump housing 21 so as to be in contact with the first casing 41 on the internal side. A recess 44 is formed in the second casing 42 on a side facing the first casing 41. The impeller 23 is rotatably housed in the recess 44.

(38) The first casing 41 has a suction hole 45 passing through the first casing 41 along the axial direction of the pump housing 21. The opening of the suction hole 45 facing the second casing 42 is formed so as to face a part of the blade grooves 37 of the impeller 23.

(39) The second casing 42 includes a discharge hole 46 passing through the second casing 42 along the axial direction of the pump housing 21. The opening of the discharge hole 46 facing the first casing 41 is formed so as to face a part of the blade grooves 37 of the impeller 23. The discharge hole 46 is disposed so as not to face the suction hole 45 along the axial direction of the pump housing 21.

(40) The first casing 41 has a first flow groove 47 for allowing fuel to flow on a surface thereof facing the second casing 42. The first flow groove 47 is formed in an annular shape so as to face a part of the blade grooves 37 of the impeller 23. The first end of the annular first flow groove 47 faces the suction hole 45, and the second end of the first flow groove 47 faces the discharge hole 46.

(41) In the recess 44 of the second casing 42, a second flow groove 48 for allowing fuel to flow is formed on the surface facing the first casing 41. The second flow groove 48 is formed in an annular shape so as to face a part of the blade grooves 37 of the impeller 23. The first end of the annular second flow groove 48 faces the suction hole 45, and the second end of the second flow groove 48 faces the discharge hole 46.

(42) When the impeller 23 rotates and the fuel is pumped up from the suction hole 45, the fuel flows through a fuel flow path formed by the first flow groove 47, the second flow groove 48 and the blade groove 37. When the fuel reaches the second ends of the first flow groove 47 and the second flow groove 48, the fuel is discharged from the discharge hole 46.

(43) The motor cover 25 fixes the pump motor 22 in the pump housing 21. The motor cover 25 is installed so as to close the opening of the pump housing 21 at the second end of the pump housing 21 formed in a cylindrical shape along the cylindrical axial direction.

(44) The motor cover 25 includes a discharge hole 51 passing through the motor cover 25 along the axial direction of the pump housing 21. The fuel discharged from the discharge hole 46 of the pump case 24 is guided to the discharge hole 51 of the motor cover 25 through a fuel passage 53 formed between the rotor 31 of the pump motor 22 and the stator 32. Then, the fuel guided to the discharge hole 51 is discharged from the discharge hole 51 to the outside of the fuel pump 3.

(45) Next, a procedure of a motor control process executed by the CPU 13a of the control unit 13 will be described. The motor control process is repeatedly executed during the operation of the control unit 13. The motor control process is terminated when a command to stop driving the pump motor 22 is received from the engine controller 7.

(46) When the motor control process is executed, as shown in FIG. 4, the CPU 13a determines in S10 whether a command to start driving the pump motor 22 has been received from the engine controller 7. When the command for instructing the start of driving is not received, the CPU 13a ends the motor control process.

(47) When the command instructing the start of driving is received, the CPU 13a executes a rotor positioning control in S20. Specifically, the CPU 13a sets the rotational position of the rotor 31 at a predetermined reference angle by energizing a stator coil of a specific phase (for example, between U and V) preset for initial driving of the pump motor 22 via the inverter circuit 11.

(48) Next, in **S30**, the CPU **13a** performs a feedback control so that the motor rotation speed and the target rotation speed coincide with each other. In the present embodiment, the CPU **13a** executes PI control as the feedback control. Specifically, the CPU **13a** calculates the duty of the PWM control based on a feedback control amount obtained by adding a value obtained by multiplying a deviation between the motor rotation speed and the target rotation speed by a proportional gain and a value obtained by multiplying an integral value of the deviation by an integral gain. Then, the CPU **13a** selects two switching elements to be turned on in synchronization with the rotational position of the pump motor **22**. Further the CPU **13a** maintains one of the selected two switching elements in the ON state, and periodically switches the other switching element between the ON state and the OFF state in accordance with the duty.

(49) In **S40**, the CPU **13a** determines whether the motor control is normal. Specifically, the CPU **13a** determines that the motor control is normal when the rotational position of the pump motor **22** corresponds to the current energization pattern. When the rotational position of the pump motor **22** does not correspond to the current energization pattern, the CPU **13a** determines that the motor control is not normal.

(50) The control unit **13** controls the pump motor **22** by sequentially switching the first energization pattern, the second energization pattern, the third energization pattern, the fourth energization pattern, the fifth energization pattern, and the sixth energization pattern in order starting with the earliest.

(51) For example, the first energization pattern turns on the U-phase high-side switch and the V-phase low-side switch. The second energization pattern turns on the V-phase high-side switch and the W-phase low-side switch. The third energization pattern turns on the V-phase high-side switch and the U-phase low-side switch. The fourth energization pattern turns on the U-phase high-side switch and the W-phase low-side switch. The fifth energization pattern turns on the W-phase high-side switch and the U-phase low-side switch. The sixth energization pattern turns on the W-phase high side switch and the V phase low side switch.

(52) When it is determined in **S40** that the motor control is normal, the CPU **13a** proceeds to **S30**. When it is determined in **S40** that the motor control is not normal, the CPU **13a** increments the number of control failures COUNT_F in **S50**.

(53) In **S60**, the CPU **13a** determines whether the number of control failures COUNT_F is larger than a preset abnormality determination value **J1** (for example, 10 times). When the number of control failures COUNT_F is equal to or smaller than the abnormality determination value **J1**, the CPU **13a** proceeds to **S20**. When the number of control failures COUNT_F is larger than the abnormality determination value **J1**, the CPU **13a** executes an abnormality check of the fuel pump controller **8** in **S70**. For example, the CPU **13a** checks whether a short circuit or disconnection has occurred in the wiring between the fuel pump controller **8** and the pump motor **22**, or whether a short circuit or disconnection has occurred in the wiring inside the fuel pump controller **8**.

(54) In **S80**, the CPU **13a** determines whether an abnormality has occurred in the fuel pump controller **8** based on the check result in **S70**. When an abnormality has occurred in the fuel pump controller **8**, the CPU **13a** ends the motor control process.

(55) When an abnormality has not occurred in the fuel pump controller **8**, the CPU **13a** executes a torque abnormality check in **S90**. Specifically, the CPU **13a** first sets the target rotation speed to the first check target rotation speed set in advance for the torque abnormality check, sets the start duty to the first check start duty set in advance for the torque abnormality check, and starts the pump motor **22**. The first check target rotation speed is set to be higher than the target rotation speed when the pump motor **22** is started in a normal state. The first check start duty is set to be smaller than the start duty when the pump motor **22** is started in a normal state.

(56) The first check target rotation speed and the first check start duty are conditions for making the start of the pump motor **22** likely to fail. If the torque is large when the impeller **23** is stationary, it is necessary to increase the force for operating the impeller **23**. However, when the force for

operating the impeller **23** is increased, the acceleration at the time when the impeller **23** starts to move becomes larger than the normal time. In this case, the difference between the acceleration assumed at the time of designing the fuel pump controller **8** and the acceleration at the time when the pump motor **22** rotates becomes too large. Since the **0** cross of the induced voltage of the pump motor **22** is covered with the mask for restricting erroneous detection, the pump motor **22** is out of phase. Therefore, when the first check target rotation speed is high, the start of the pump motor **22** can be easily failed.

(57) If the torque is large when the impeller **23** is stationary, the impeller **23** cannot be set to the prescribed position at the time of starting the pump motor **22**. In this case, the pump motor **22** cannot be started satisfactorily. Therefore, when the starting duty is small, the start of the pump motor **22** can be easily failed.

(58) Then, the CPU **13a** determines whether the start of the pump motor **22** is successful after starting the pump motor **22** at the first check target rotation speed and the first check start duty.

(59) Next, the CPU **13a** sets the target rotation speed to the second check target rotation speed set in advance for the torque abnormality check, sets the start duty to the second check start duty set in advance for the torque abnormality check, and starts the pump motor **22**. The second check target rotation speed is set to be lower than the target rotation speed when the pump motor **22** is started in a normal state. The second check start duty is set to be larger than the start duty when the pump motor **22** is started in a normal state. The second check target rotation speed and the second check start duty are conditions for making it easy to successfully start the pump motor **22**.

(60) Then, the CPU **13a** determines whether the start of the pump motor **22** is successful after starting the pump motor **22** at the second check target rotation speed and the second check start duty.

(61) When the torque abnormality check ends, in **S100**, the CPU **13a** determines whether a torque abnormality has occurred based on the check result in **S90**. Specifically, the CPU **13a** determines that the torque abnormality has occurred when the start of the pump motor **22** has failed at the first check target rotation speed and the first check start duty and the start of the pump motor **22** has succeeded at the second check target rotation speed and the second check start duty.

(62) If the torque abnormality has not occurred, the CPU **13a** proceeds to **S20**. When the torque abnormality has occurred, the CPU **13a** transmits a torque abnormality notification indicating that the torque abnormality has occurred to the engine controller **7** in **S110**. The engine controller **7** that has received the torque abnormality notification transmits the torque abnormality notification to a meter control device that controls a meter panel displaying the vehicle state and the like to the driver. The meter control device that has received the torque abnormality notification causes the meter panel to display that the torque abnormality has occurred. Accordingly, the driver of the vehicle can recognize that the torque abnormality occurs in the fuel pump **3**.

(63) Further, in **S120**, the CPU **13a** changes a control parameter for controlling the pump motor **22**. Specifically, the CPU **13a** executes at least one of a first parameter change, a second parameter change, and a third parameter change, which will be described later. Accordingly, even when the torque abnormality occurs, it is possible to continue the rotation of the pump motor **22** and continue the pumping of the fuel by the fuel pump **3**.

(64) The first parameter change is to change the proportional gain and the integral gain of the feedback control to be smaller than the gain when the pump motor **22** is driven in the normal state.

(65) The second parameter change is to change the target rotation speed of the feedback control to be smaller than the target rotation speed when the pump motor **22** is driven in the normal state. The third parameter change is to change the starting duty so as to be larger than the starting duty when the pump motor **22** is started in a normal state.

(66) In **S130**, the CPU **13a** limits the upper limit value of the current flowing through the pump motor **22** (hereinafter, the upper limit motor current value) and the upper limit value of the motor rotation speed of the pump motor **22** (hereinafter, the upper limit motor rotation speed), and

proceeds to S20. Specifically, the CPU 13a sets the upper limit motor current value of the pump motor 22 to the torque abnormality-time upper limit current value set to be smaller than the value when the pump motor 22 is driven in the normal state. In addition, the CPU 13a sets the upper limit motor rotation speed of the pump motor 22 to the torque abnormality-time upper limit rotation speed set to be smaller than the value when the pump motor 22 is driven in the normal state.

(67) The fuel pump controller 8 determines whether the motor rotation control for rotating the pump motor 22 has failed. Then, the fuel pump controller 8 calculates the number of control failures COUNT_F. Further, the fuel pump controller 8 determines whether an abnormality has occurred in the pump motor 22 based on the number of control failures COUNT_F. This abnormality means that the torque applied to the impeller 23, which is fixed to the pump motor 22 and rotates by the driving of the pump motor 22, increases. The number of control failures COUNT_F is a parameter having a positive correlation with the frequency of failures of the motor rotation control. The expression “having a positive correlation with the frequency” includes not only a case where the parameter increases in a stepwise manner as the frequency increases, but also a case where the parameter continuously increases as the frequency increases.

(68) When it is determined that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 controls the pump motor 22 by changing a motor control parameter for performing the motor rotation control so as to suppress the rotation of the pump motor 22 from being stopped.

(69) The fuel pump controller 8 can determine that an abnormality has occurred when the frequency of failures in the motor rotation control increases, in which foreign matter is caught in the impeller 23 of the pump 3 or the impeller 23 of the fuel pump 3 interferes with the first casing 41 or the second casing 42.

(70) Therefore, the fuel pump controller 8 can take appropriate measures to restrict the rotation of the pump motor 22 from being stopped when the abnormality occurs. Accordingly, the fuel pump controller 8 can suppress the rotation of the pump motor 22 from being stopped when the abnormality occurs, and can improve the reliability of the fuel pump 3.

(71) When the fuel pump controller 8 determines that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 changes the motor control parameter so that the increasing speed of the motor application voltage applied to the pump motor 22 is lower than that in a normal state. Specifically, the fuel pump controller 8 changes the proportional gain and the integral gain of the feedback control to be smaller than the values when the pump motor 22 is driven in the normal state.

(72) When the fuel pump controller 8 determines that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 controls the pump motor 22 by setting the target rotation speed of the feedback control to be smaller than that in the normal state.

(73) The fuel pump controller 8 changes the motor control parameter so that the motor application torque applied to the pump motor 22 at the time of starting the pump motor 22 becomes larger than that at the normal time. Specifically, the fuel pump controller 8 changes the start duty so as to be larger than a value when the pump motor 22 is driven in a normal state.

(74) When it is determined that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 limits the value of the motor current flowing through the pump motor 22 to a value equal to or less than the torque abnormality-time upper limit current value, which is an upper limit value set to be smaller than the normal time. Accordingly, the fuel pump controller 8 can suppress the occurrence of a situation in which an excessive current flows through the pump motor 22 and the pump motor 22 is damaged when an abnormality occurs in the pump motor 22, and can continue the driving of the fuel pump 3.

(75) When it is determined that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 notifies the engine controller 7 that an abnormality has occurred in the pump motor 22 by transmitting a torque abnormality notification to the engine controller 7. Accordingly, when an abnormality occurs in the pump motor 22, the fuel pump controller 8 can cause the engine

controller 7 to execute a process for coping with the abnormality or cause the driver of the vehicle to recognize the occurrence of the abnormality.

(76) When the fuel pump controller 8 determines that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 limits the rotation speed of the pump motor 22 to the torque abnormality-time upper limit rotation speed or less, which is an upper limit value set to be smaller than that in a normal state. Accordingly, the fuel pump controller 8 can suppress the occurrence of a situation in which an excessive current flows through the pump motor 22 and the pump motor 22 is damaged when an abnormality occurs in the pump motor 22, and can continue the driving of the fuel pump 3.

(77) In the embodiment, the fuel pump controller 8 corresponds to a motor control device, and the pump motor 22 corresponds to a motor. In addition, S40 corresponds to a rotation control determination unit, S50 corresponds to a parameter calculation unit, the number of control failures COUNT_F corresponds to a control failure frequency parameter, S60 and S100 correspond to an abnormality determination unit, and S120 corresponds to a stop suppression unit.

(78) S130 corresponds to a current limiting unit, the torque abnormality-time upper limit current value corresponds to an abnormality-time upper limit, S110 corresponds to an abnormality notification unit, S130 corresponds to a rotation speed limiting unit, and the torque abnormality-time upper limit rotation speed corresponds to an abnormality-time upper limit.

Second Embodiment

(79) A second embodiment will be described with reference to the drawings. Note that in the second embodiment, portions different from the first embodiment are described. Common configurations are denoted by the same reference numerals.

(80) The fuel supply system 1 of the second embodiment is different from the fuel supply system of the first embodiment in that the motor control process is changed. The motor control process of the second embodiment is different from that of the first embodiment in that the processes of S60 and S110 are omitted and the processes of S15, S55, S65, and S115 are added.

(81) That is, as shown in FIG. 5, when a command instructing the start of driving is received in S10, the CPU 13a increments the number of starts COUNT_S in S15, and proceeds to S20.

(82) When the process of S50 ends, the CPU 13a calculates the control failure probability PROB_F by dividing the number of control failures COUNT_F by the number of starts COUNT_S in S55.

(83) Then, in S65, the CPU 13a determines whether the control failure probability PROB_F is larger than a preset abnormality determination value J2. If the control failure probability PROB_F is equal to or less than the abnormality determination value J2, the CPU 13a proceeds to S15. When the control failure probability PROB_F is larger than the abnormality determination value J2, the CPU 13a proceeds to S70.

(84) When it is determined in S100 that the torque abnormality has occurred, the CPU 13a transmits control failure probability information indicating the value of the control failure probability PROB_F to the engine controller 7 in S115, and the process proceeds to S120.

(85) The fuel pump controller 8 determines whether the motor rotation control for rotating the pump motor 22 has failed. Then, the fuel pump controller 8 calculates the control failure probability PROB_F. Further, the fuel pump controller 8 determines whether an abnormality has occurred in the pump motor 22 based on the control failure probability PROB_F. The control failure probability PROB_F is a parameter having a positive correlation with the frequency of failures in the motor rotation control.

(86) When it is determined that an abnormality has occurred in the pump motor 22, the fuel pump controller 8 controls the pump motor 22 by changing a motor control parameter for performing the motor rotation control so as to suppress the rotation of the pump motor 22 from being stopped.

(87) When the value of the control failure probability PROB_F increases, the fuel pump controller 8 can determine that an abnormality has occurred in which foreign matter is caught in the impeller 23 of the fuel pump 3 or the impeller 23 of the fuel pump 3 interferes with the first casings 41 or

the second casing **42**.

(88) Therefore, the fuel pump controller **8** can take appropriate measures to restrict the rotation of the pump motor **22** from being stopped when the abnormality occurs. Accordingly, the fuel pump controller **8** can suppress the rotation of the pump motor **22** from being stopped when the abnormality occurs, and can improve the reliability of the fuel pump **3**.

(89) In the embodiment, **S15**, **S50**, and **S55** correspond to a parameter calculation unit and a failure probability calculation unit, the control failure probability **PROB_F** corresponds to a control failure frequency parameter, and **S65** and **S100** correspond to an abnormality determination unit.

(90) Although the embodiments of the present disclosure has been described above, the present disclosure is not limited to the above embodiment, and various modifications can be made.

First Modification

(91) For example, in the above embodiment, the torque abnormality check is executed when the number of control failures **COUNT_F** is larger than the abnormality determination value **J1**, and it is determined whether an abnormality has occurred in the pump motor **22** based on the result of the torque abnormality check. However, when the number of control failures **COUNT_F** is larger than the abnormality determination value **J1**, it may be determined that an abnormality has occurred in the pump motor **22**.

Second Modification

(92) In the above embodiment, after the process of starting the pump motor **22** at the first check target rotation speed and the first check start duty is performed, the process of starting the pump motor **22** at the second check target rotation speed and the second check start duty is performed. However, the determination may be made without executing the process of starting the pump motor **22** at the second check target rotation speed and the second check start duty. That is, the CPU **13a** may determine that the torque abnormality has occurred when the start of the pump motor **22** has failed as a result of performing the process of starting the pump motor **22** at the first check target rotation speed and the first check start duty.

Third Modification

(93) In the embodiment, the number of control failures **COUNT_F** or the control failure probability **PROB_F** having a positive correlation with the frequency of failures in the motor rotation control is calculated, and it is determined whether an abnormality has occurred in the pump motor **22** based on the number of control failures **COUNT_F** or the control failure probability **PROB_F**. However, the number of control successes or the control success probability having a negative correlation with the frequency of failures in the motor rotation control may be calculated, and it may be determined whether an abnormality has occurred in the pump motor **22** based on the number of control successes or the control success probability.

(94) The control unit **13** and the techniques thereof according to the present disclosure may be implemented by one or more special-purposed computers. Such a special-purposed computer may be provided (i) by configuring (a) a processor and a memory programmed to execute one or more functions embodied by a computer program, or (ii) by configuring (b) a processor including one or more dedicated hardware logic circuits, or (iii) by configuring by a combination of (a) a processor and a memory programmed to execute one or more functions embodied by a computer program and (b) a processor including one or more dedicated hardware logic circuits.

(95) The computer program may store a computer-readable non-transitional tangible recording medium as an instruction to be executed by the computer. The technique for realizing the functions of each unit included in the control unit **13** does not necessarily need to include software, and all the functions may be realized using one or a plurality of hardware circuits.

(96) The multiple functions of one component in the above embodiment may be realized by multiple components, or a function of one component may be realized by the multiple components. In addition, multiple functions of multiple components may be realized by one component, or a single function realized by multiple components may be realized by one component. Moreover,

part of the configuration of the above embodiment may be omitted. At least a part of the configuration of the above embodiment may be added to or replaced with the configuration of another embodiment.

(97) In addition to the fuel pump controller **8**, the present disclosure may be implemented in various forms such as a system including the fuel pump controller **8** as a component, a program for causing a computer to function as the fuel pump controller **8**, a non-transitory tangible recording medium such as a semiconductor memory storing the program, and a motor control method.

Claims

1. A motor control device configured to control a motor, comprising: a microcontroller configured to determine whether a motor rotation control to rotate the motor fails; configured to calculate a control failure frequency parameter having a correlation with a frequency of failure in the motor rotation control based on a determination result by the microcontroller; configured to calculate a control failure probability that is a probability of failure in the motor rotation control by dividing a number of control failures by a number of starts; configured to determine whether an abnormality has occurred in the motor based on the control failure probability; and configured to suppress stop of rotation of the motor by changing a motor control parameter to perform the motor rotation control when the microcontroller determines that the abnormality has occurred in the motor.
2. The motor control device according to claim 1, wherein the microcontroller changes the motor control parameter such that an increasing speed of a motor application voltage applied to the motor is lower than that at a normal time.
3. The motor control device according to claim 1, wherein the microcontroller is further configured to perform a feedback control of the number of rotations of the motor, and the microcontroller controls the motor by setting a target number of rotations of the feedback control to be smaller than that at a normal time.
4. The motor control device according to claim 1, wherein the microcontroller changes the motor control parameter such that a motor application torque applied to the motor when the motor is started is larger than that at a normal time.
5. The motor control device according to claim 1, wherein the microcontroller is further configured to limit a value of a motor current flowing through the motor, when the microcontroller determines that an the abnormality has occurred in the motor, to be equal to or less than the abnormality-time upper limit set to be smaller than that at a normal time.
6. The motor control device according to claim 1, wherein the microcontroller is further configured to notify that the abnormality has occurred in the motor when the microcontroller determines that the abnormality has occurred in the motor.
7. The motor control device according to claim 1, wherein the microcontroller is further configured to limit a rotation speed of the motor, when the microcontroller determines that an the abnormality has occurred in the motor, to be smaller than or equal to the abnormality-time upper limit set to be smaller than that at a normal time.
8. The motor control device according to claim 1, wherein the microcontroller is further configured to execute a first parameter change, as a change of the motor control parameter, when a start of the motor fails, the first parameter change is to change a proportional gain and an integral gain of a feedback control of the number of rotations of the motor to be smaller than those when the motor is driven in a normal state, and the microcontroller is further configured to maintain rotation of the motor by executing the first parameter change, even when the control failure probability is more than an abnormality determination value.
9. The motor control device according to claim 1, wherein the microcomputer is further configured to execute a second parameter change, as a change of the motor control parameter, when a start of the motor fails, the second parameter change is to change a target rotation speed of a feedback

control of the number of rotations of the motor to be smaller than that when the motor is driven in the normal state, and the microcomputer maintains rotation of the motor by executing the second parameter change, even when the control failure probability is more than an abnormality determination value.

10. The motor control device according to claim 1, wherein the microcomputer is further configured to execute a third parameter change, as a change of the motor control parameter, when a start of the motor fails, the third parameter change is to change a starting duty of a feedback control of the number of rotations of the motor to be larger than that when the motor is started in a normal state, and the microcomputer maintains rotation of the motor by executing the third parameter change, even when the control failure probability is more than an abnormality determination value.

11. A motor control device configured to control a motor, comprising: a computer including a processor and a memory that stores instructions configured to, when executed by the processor, cause the processor to determine whether a motor rotation control to rotate the motor fails; calculate a control failure frequency parameter having a correlation with a frequency of failure in the motor rotation control based on a result of the determining; calculate a control failure probability that is a probability of failure in the motor rotation control by dividing a number of control failures by a number of starts determine whether an abnormality has occurred in the motor based on the control failure probability; and suppress stop of rotation of the motor by changing a motor control parameter to perform the motor rotation control when it is determined that the abnormality has occurred in the motor.

12. The motor control device according to claim 11, wherein the memory stores further instructions configured to, when executed by the processor, cause the processor to execute a first parameter change, as a change of the motor control parameter, when a start of the motor fails, the first parameter change is to change a proportional gain and an integral gain of a feedback control of the number of rotations of the motor to be smaller than those when the motor is driven in a normal state, and the memory stores further instructions configured to, when executed by the processor, cause the processor to maintain rotation of the motor by executing the first parameter change, even when the control failure probability is more than an abnormality determination value.

13. The motor control device according to claim 11, wherein the memory stores further instructions configured to, when executed by the processor, cause the processor to execute a second parameter change, as a change of the motor control parameter, when a start of the motor fails, the second parameter change is to change a target rotation speed of a feedback control of the number of rotations of the motor to be smaller than that when the motor is driven in the normal state, and the memory stores further instructions configured to, when executed by the processor, cause the processor to maintain rotation of the motor by executing the second parameter change, even when the control failure probability is more than an abnormality determination value.

14. The motor control device according to claim 11, wherein the memory stores further instructions configured to, when executed by the processor, cause the processor to execute a third parameter change, as a change of the motor control parameter, when a start of the motor fails, the third parameter change is to change a starting duty of a feedback control of the number of rotations of the motor to be larger than that when the motor is started in a normal state, and the memory stores further instructions configured to, when executed by the processor, cause the processor to maintain rotation of the motor by executing the third parameter change, even when the control failure probability is more than an abnormality determination value.
